

Python Charts and Their Analytical Use Cases

Architectural design and decision-support rationale for exploratory churn visualizations in `churn_analysis.ipynb`.

The Python diagnostic script processes the customer dataset using `pandas` and `matplotlib` to translate raw numerical records into targeted visual intuition. The reporting suite is arranged into a combined **2×2 grid (Plots 1–4)** for immediate segment diagnostics, followed by a dedicated **correlation driver chart (Plot 5)** for executive review.

PART 1 The 2×2 Exploratory Visual Grid

Plot 1: Overall Churn

BAR CHART

PRIMARY QUESTION:

What share of customers is leaving the bank?

ANALYTICAL USE CASE:

Establishes the empirical baseline attrition rate (~20.4%) and highlights class imbalance in the target column. Essential before training classification models, as a baseline model could naïvely achieve ~80% accuracy by predicting "stays" exclusively.

Total Portfolio (10,000 Records)



Plot 2: Churn by Geography

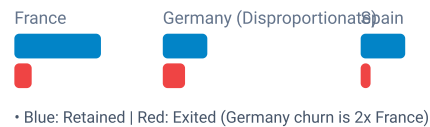
GROUPED BAR

PRIMARY QUESTION:

Where is customer churn geographically concentrated?

ANALYTICAL USE CASE:

Geographic segment analysis. When one region's Exited bar is disproportionately high versus its Not Exited volume (e.g., Germany at ~32.4% vs. France/Spain at ~16%), it signals localized operational friction, local competitors, or regional pricing disparities rather than global flaws.



Plot 3: Age Distribution

OVERLAPPING HISTOGRAMS

PRIMARY QUESTION:

Which demographic age groups are leaving the bank?

ANALYTICAL USE CASE:

Demographic profiling. If leavers are older (peaking between ages 45–60), retention tactics must pivot toward wealth preservation, retirement products, and high-touch relationship managers rather than simple mobile gamification or onboarding app flows.

Plot 4: Account Balance

BOX PLOT

PRIMARY QUESTION:

Are we losing high-value clients or dormant zero balances?

ANALYTICAL USE CASE:

Quantifies immediate capital risk. An elevated median and interquartile box for Exited accounts proves that departing clients hold more deposits than retained ones, making them the primary priority for personalized outreach by senior branch relationship managers.

PART 2 Plot 5: Correlation of Features with Churn

While Plots 1–4 isolate individual categorical breakdowns, **Plot 5** aggregates the mathematical relationship between all numerical and encoded attributes against the target label (Exited) using Pearson correlation coefficients.

Plot 5: Feature Correlation Vector

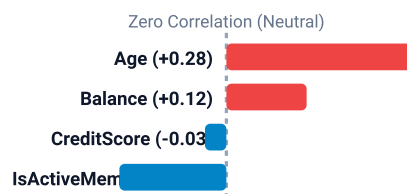
HORIZONTAL BAR CHART

QUESTION IT ANSWERS:

Which quantifiable behavioral and account features exert the strongest directional pull on customer churn?

EXECUTIVE USE CASE:

Serves as a driver ranking for executive sponsors and retention strategists. Bars extending to the right represent factors that correlate with increased churn (e.g., Age), while bars extending to the left represent protective factors that correlate with customer retention (e.g., IsActiveMember).



Methodological Note: Always strip artificial identifier columns (such as RowNumber and CustomerId) before computing coefficients to avoid spurious mathematical noise. Correlation reflects linear co-movement and does not prove direct causation.

Summary Matrix: Chart Utility by Operational Audience

Chart	Target Question	Core Business Value	Primary Beneficiary
Plot 1: Overall	Portfolio retention share?	Sets baseline performance & guards against accuracy paradoxes.	Data Science & Risk Modeling
Plot 2: Geography	Where is churn worst?	Detects local market dysfunction vs. firm-wide system problems.	Regional Branch Leadership
Plot 3: Age	Which age cohorts leave?	Drives demographic product alignment (wealth vs. onboarding).	Product & Lifecycle Marketing
Plot 4: Balance	Are high-value clients lost?	Determines balance at risk to prioritize relationship manager calls.	Private Wealth & Concierge Desk
Plot 5: Correlation	What factors drive churn?	Ranks directional importance of all variables simultaneously.	Executive Steering Committee